

Background

Ashwagandha is a shrub that has been utilized as a traditional medicine to reduce stress in patients. Recent studies investigated the therapeutic effect of the Ashwagandha and suggested that the **stress-relieving effects** occur from its moderation of stress responses, as well as serving as an agent for neuroimmunomodulation (Lopresti et al., 2019). Pharmacological studies have also demonstrated it to have various neuroprotective effects, as well as being a useful anti-inflammatory, antitumor, antimicrobial, cardio-protective, and antioxidative properties. No adverse effects were recorded because of its administration, and toxicity studies showed the compound as safe (Mariano et al., 2020; Mandilik & Namdeo, 2021).

Parkinson's Disease (PD) is a progressive movement disorder which emerges from neurological issues in the Brain. Currently PD has no cure, and a substantial portion of contemporary research prioritizes the treatment of symptoms, which include a **decline in memory function**, from stiffness and tremors to **difficulty in movement** and communication, as well as changes in blood pressure, cognitive issues, pain, and fatigue (Gibson et al., 2021; National Institute of Neurological Disorders and Stroke, n.d.).

The heat shock response (HSR) is a mechanism used by the body to manage cellular stress, such as stress resulting from Parkinson's Disease. The HSR utilizes heat shock factors (HSFs) to regulate itself, HSFs play an essential role in the activation of Heat shock proteins (HSPs), which is used as a chaperone to treat damage resulting from oxidative stress (Kurop et al., 2021). Because of PD's relation to oxidative stress, antioxidants are partially able to treat its symptoms.

In a similar experiment conducted last year, Ashwagandha was found to be successful in improving the symptoms of PD, however because Ashwagandha is an antioxidant, no definitive answer on the efficacy of improving the HSR could be found, as any improvements could have come because of mere antioxidant administration.

Data

Average Times in Negative Geotaxis Assays for <i>Drosophila melanogaster</i> to climb 5.4cm (s), 15 Trials		
Control Flies		
Untreated	Vitamin C	Ashwagandha
6.87	12.5	7.62
PD Afflicted Flies		
Untreated	Vitamin C	Ashwagandha
6.47	8.31	11.83

Table 1: Average times for 15 groups of *Drosophila melanogaster* in negative geotaxis assays to climb 5.4cm.

Distribution of Flies between Illuminated and Dark Vials						
Trial	Before Quinine Placement		During Quinine Placement		After Quinine Removal	
	# In Light Vial	# In Dark Vial	# In Light Vial	# In Dark Vial	# In Light Vial	# In Dark Vial
Unafflicted/Untreated	3	2	1	4	4	1
Unafflicted/Vitamin C	4	1	1	4	4	1
Unafflicted/Ashwagandha	4	1	1	4	3	2
Parkinson's/Untreated	4	1	2	3	4	1
Parkinson's/Vitamin C	4	1	2	3	4	1
Parkinson's/Ashwagandha	3	2	1	4	3	2

Table 2: The distributions of *Drosophila melanogaster* in illuminated and unilluminated vials before, during, and after quinine insertion into the illuminated vials, sorted by type of treatment and affliction.

Investigating the effect of Ashwagandha on the Heat Shock Response of *Drosophila melanogaster*

TMED- Translational Medicine

Methodology

Equipment

- Plastic *Drosophila melanogaster* vials with flugs
- Video Camera
- Vitamin C & Ashwagandha
- Quinine
- 70% Ethanol
- Choice Chamber

Introduction

Hypothesis:

PD-afflicted *Drosophila melanogaster* will exhibit better motor and memory function if treated with Ashwagandha over other antioxidants or absence of treatment.

Independent Variable:
Type of dietary supplement ingested by *Drosophila melanogaster*

Dependent Variable:
Memory and motor function of *Drosophila melanogaster*

Controls:

Negative: Untreated *Drosophila*

Positive: Treated Healthy *Drosophila*

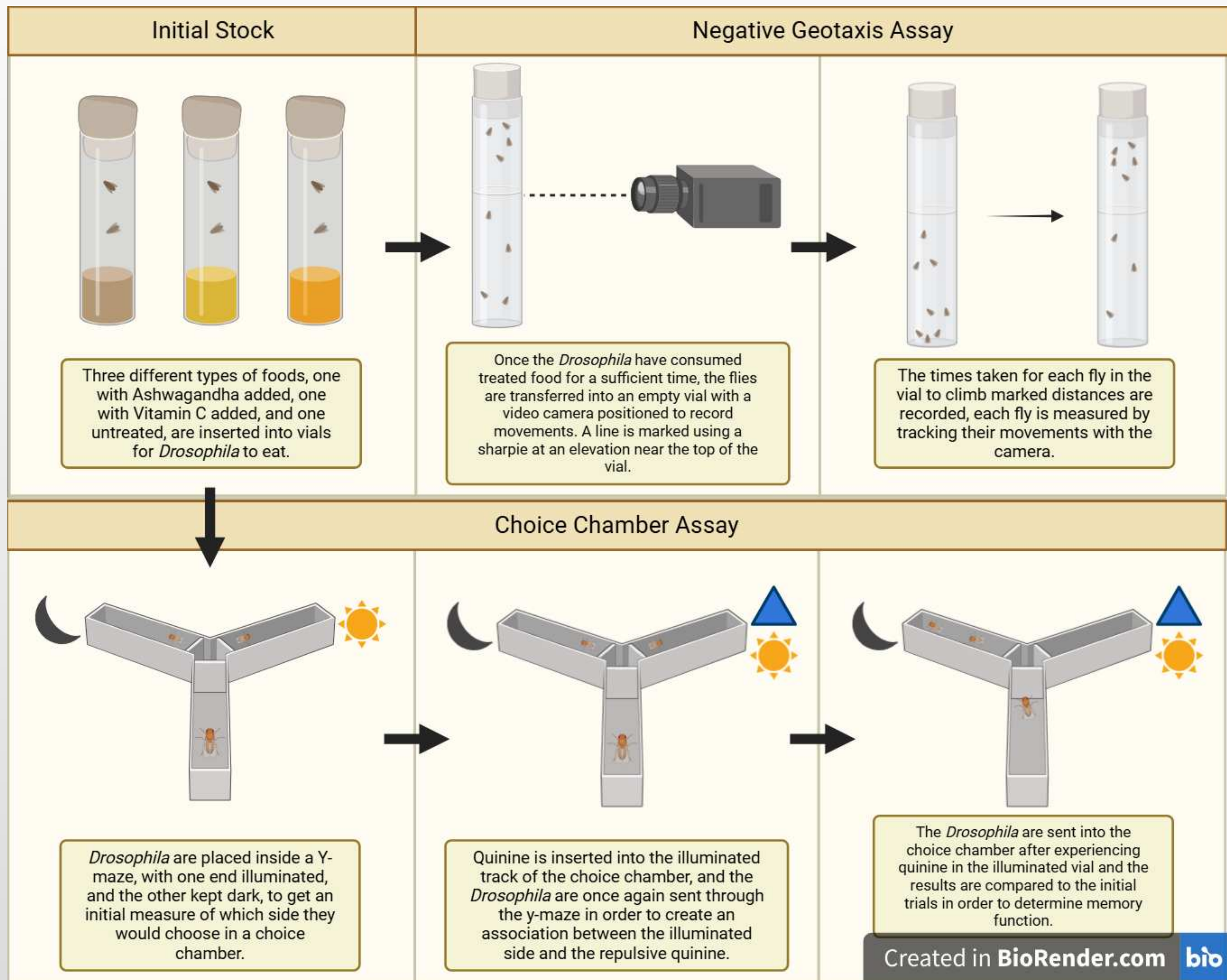


Figure 1: Description of methodologies for the negative geotaxis and choice chamber assays.

Results

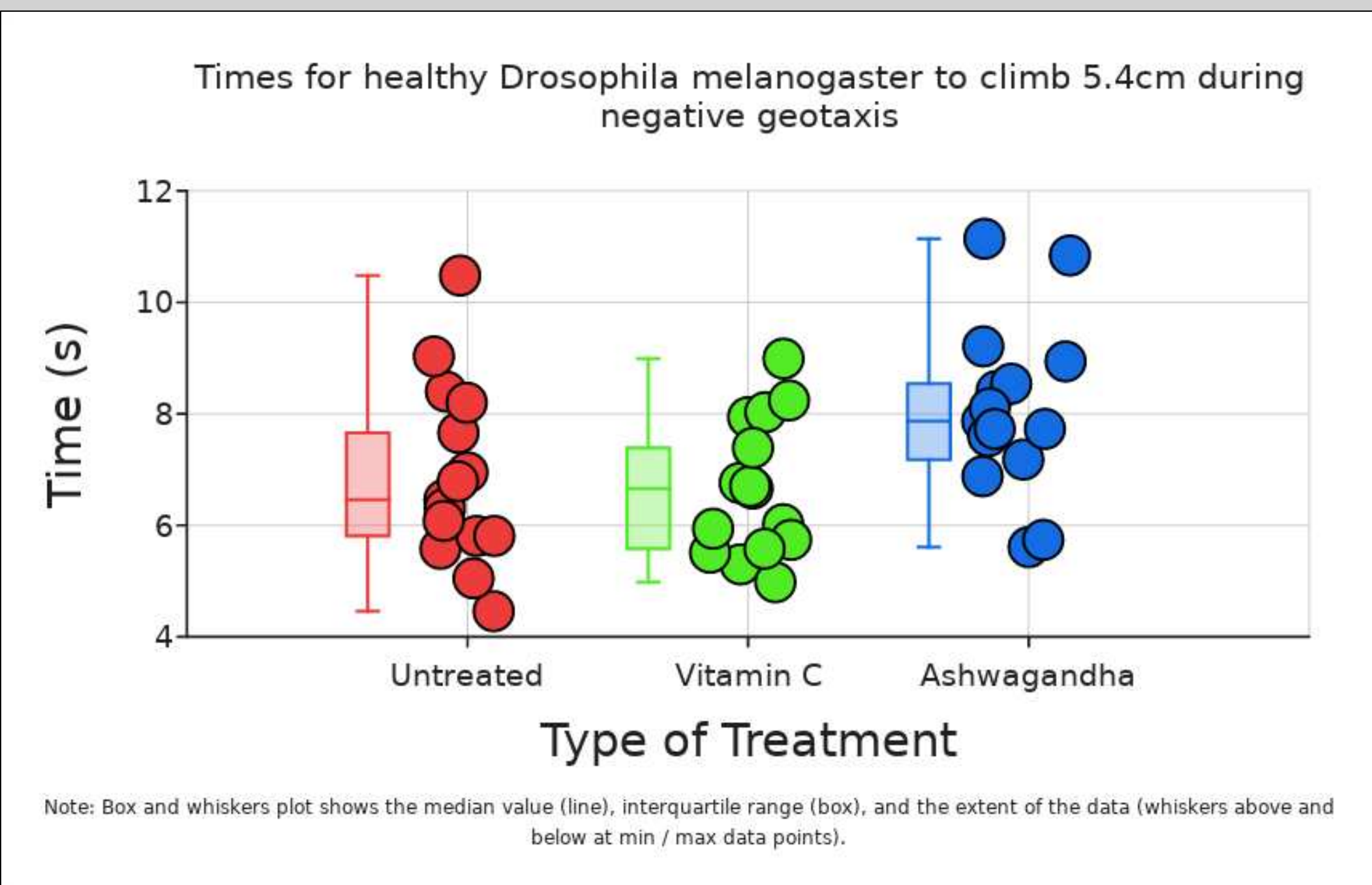


Figure 2: Times for healthy *Drosophila melanogaster* (groups of 10-12) to climb 5.4cm during negative geotaxis assays.

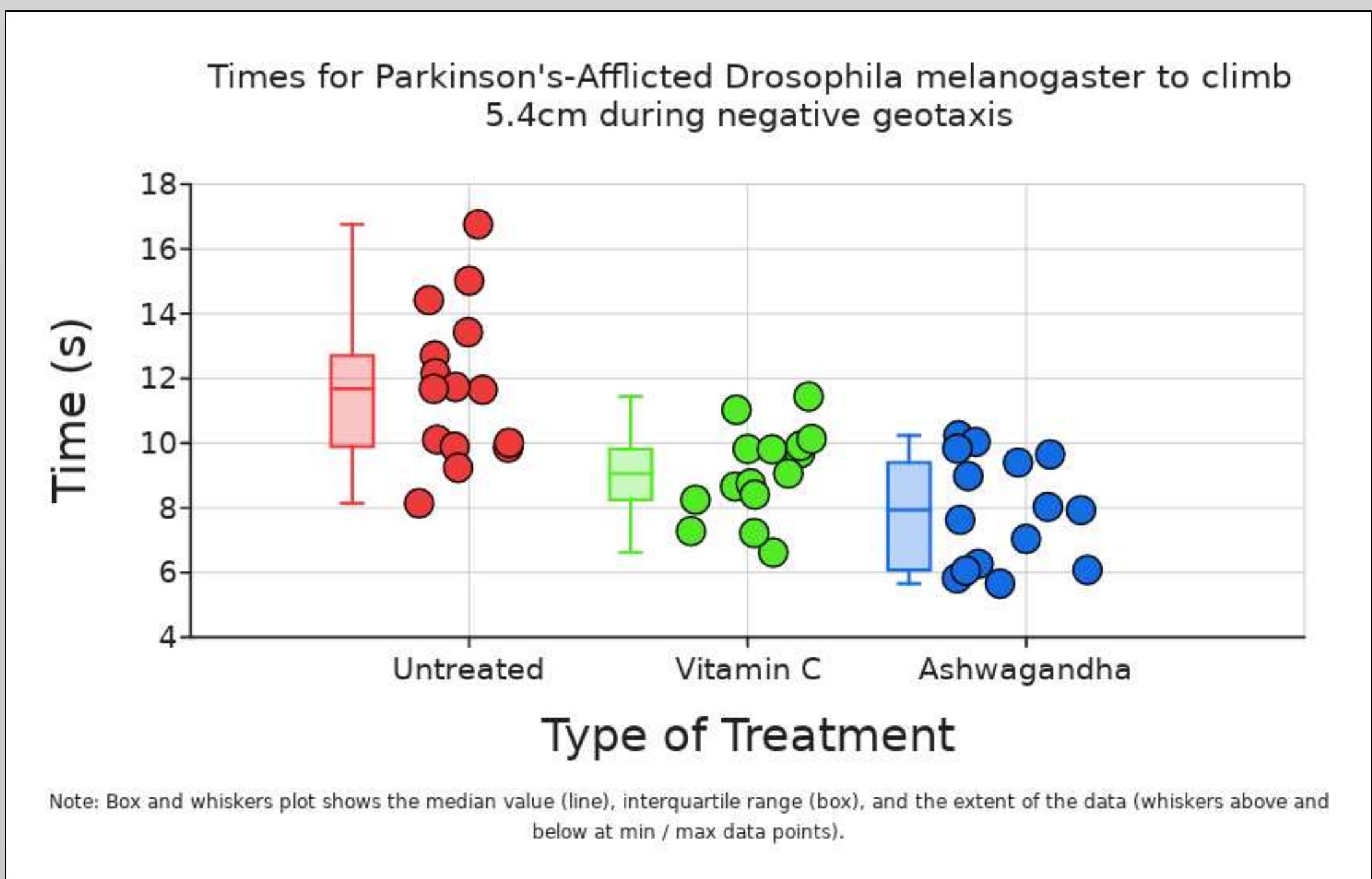


Figure 3: Times for PD afflicted *Drosophila melanogaster* (groups of 10-12) to climb 5.4cm during negative geotaxis assays.

Conclusion

After the negative geotaxis trials, a Kruskal-Wallis test demonstrated statistically significant differences in climbing time for afflicted ($p < 0.01$) and unafflicted ($p = 0.03$) groups.

- For unafflicted flies, Ashwagandha showed a significant decline in performance compared to flies treated with Vitamin C ($p = 0.04$), although this is likely due to the unintended advanced age of the flies in the Ashwagandha group during testing.
- For afflicted flies, Ashwagandha and Vitamin C delivered significant improvements in performance over untreated groups ($p = 0.01$), however between the two treatments, no significant difference was observed ($p = 0.47$).

For the behavioral assays, flies treated with vitamin C performed identically to groups without treatment, whereas ones treated with Ashwagandha demonstrated improved memory to the degree that they did just as well as healthy flies during trials with quinine present as aversive stimuli.

While ashwagandha improved motor and memory function in *Drosophila melanogaster*, the tests demonstrated that their results are no more significant than alternative antioxidants such as Vitamin C, suggesting that the benefits from Ashwagandha are derived from its antioxidative properties rather than a unique pathway with the Heat Shock Response. Regardless, a trend remained where flies treated with Ashwagandha tended to perform better on average.

References

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